# Juan Becerra

**Software Engineer – Gameplay, Animation & Tools**

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## Experience

**Electronic Arts – Lifestyle** **2025 – Present**
*Software Engineer – Tools & Tech-Art* *Kirkland, WA*

- Engineered a high-throughput Unreal Engine C++ plugin capturing and exporting runtime character trajectory, pose, and velocity state to drive motion-matching locomotion query databases.
- Architected an extensible Maya/Python animation stylization tool adopted across multiple internal studios, reducing redundant manual keyframing workflows by over 70%.
- Prototyped and integrated ML-assisted authoring pipelines into DCC tools, enabling automated asset tagging and procedural pose generation without disrupting artist control.

**Electronic Arts – Maxis Studios** **2021 – 2025**
*Software Engineer – Gameplay & Animation* *Seattle, WA (Remote)*

- Implemented core locomotion, state machines, and player-facing runtime animation systems in UE5/C++ (GAS, motion-matching, distance matching, custom IK solvers), delivering fluid, responsive character controls.
- Architected a multi-threaded asynchronous path-planning and navigation API in UE5/C++ for *Project Rene* (*The Sims*), enabling deterministic routing and real-time obstacle avoidance for shared multiplayer simulation.
- Partnered closely with gameplay designers, technical animators, and QA to prototype, iterate, and polish gameplay features while authoring in-editor tuning tools and visual debuggers.

**Lionbridge** **2020 – 2021**
*Senior Games Test Engineer* *Boise, ID*

- Led a team of QA testers performing combinatorial and regression testing for live-service AAA multiplayer titles, authoring technical repro steps and engine-level diagnostics in JIRA to accelerate developer triage.

## Projects

**Hoowan Game Engine** | *C++, OpenGL, EnTT, CMake* | GitHub **Nov. 2020**

- Engineered a custom C++ game engine featuring an OpenGL rendering submission pipeline, input event polling, and 2D collision detection and correction systems.
- Integrated an EnTT-based entity-component system (ECS) and constructed a companion desktop editor application for rapid scene composition and entity management.

**DoomSlop** | *Three.js, Vite, JavaScript* | Live Demo | GitHub **Sep. 2026**

- Built a 3D first-person shooter featuring responsive movement and camera controls, hitscan weapon raycasting, combat state feedback, custom canvas radar minimap, and automated CI/CD deployments.

## Technical Skills

**Languages**: C++ (Modern, Multi-threading), C#, Python, Rust, GLSL/HLSL, Java, JavaScript, SQL
**Gameplay & Systems**: Locomotion & Character Physics, Animation Runtime (IK, Motion-Matching), Async Navigation/Pathfinding
**Game Engines & APIs**: Unreal Engine 5 (Source, GAS, Motion Matching, Mass/Crowd), Unity, Godot, Vulkan, OpenGL
**Tools**: Unreal Plugin Development, Maya API, PySide/Qt, DCC Pipelines, PyTorch/ONNX
**Infrastructure**: Perforce (P4), Git, JIRA, Visual Studio, CMake, Docker, Linux, AWS, CI/CD

## Education & Continuing Studies

**Boise State University** **2017 – 2021**
*Bachelor of Science in Computer Science* *Boise, ID*

**University of Washington (Continuing Education)** **2026 – Present**
*Graduate Coursework in Machine Learning & Deep Learning (Non-Degree, In Progress)* *Seattle, WA*